

Stakeholder Conflict Analysis

Three stakeholders, three definitions of success, one system

Project	Course	Case Study	Version
HaulTrack - 1,800-truck fleet telematics	B.Tech CSE - Software Engineering & Project Management	No. 19	1.0 (baseline)

1. Purpose and elicitation

HaulTrack serves one system to three groups whose definitions of success differ. This document records what each group asked for, where the asks collide, how the collisions were ranked, and what was decided. Conflicts are **resolved by explicit decision**, not averaged: where a stakeholder lost something, the document says so.

Elicitation techniques used: structured interviews (fleet manager, insurer underwriter), a union-representative workshop plus a short driver questionnaire, document analysis of the abandoned vendor's safety-score specification, and a post-mortem of why it failed. The finding of the post-mortem is the anchor of this analysis: **the vendor penalised drivers on hill routes because the requirements never said braking must be judged against terrain. That is a requirements failure, so terrain fairness is written as a measurable requirement (Section 6), not left to the algorithm.**

2. Stakeholder register

ID	Stakeholder	Definition of success	Influence	Stance
S1	Fleet Manager	Route deviation alerts that are timely; a driver safety score usable to coach drivers and cut incidents and fuel waste	High (sponsor)	Supportive, wants broad use
S2	Drivers' Union	Written assurance the score cannot be used for arbitrary dismissal; no hidden surveillance; fair treatment on hard routes	High (can block adoption)	Sceptical
S3	Insurer	Credible, tamper-evident evidence that harsh braking has fallen, to justify lower premiums	Medium-High (funds the benefit)	Neutral, evidence-driven
S4	Drivers (1,800 trucks)	Be judged fairly, see their own data, have a way to challenge	Medium (via S2)	Anxious
S5	Legal / Data Protection	Lawful basis, minimisation, defined retention, access control	High (veto on release)	Cautious
S6	IT Operations	A pipeline that handles the load and can be run by the existing team	Medium	Neutral

3. Requirements as asked by each group

Ref	Stakeholder	Requirement as stated
F1	S1	Alert the dispatcher when a truck leaves its planned route.
F2	S1	Give every driver a safety score and let managers rank/compare drivers.
F3	S1	Track trucks continuously to protect assets and fuel.
U1	S2	The score must never be the basis for dismissal or discipline.
U2	S2	No tracking when the driver is off duty; no league tables or public shaming.
U3	S2	Drivers must see how their score is built and be able to appeal it.
U4	S2	Hill / mountain routes must not lower a driver's score.
I1	S3	Provide multi-period evidence that harsh braking decreased.
I2	S3	Evidence must be traceable to individual events and be hard to alter.
I3	S3	Keep a metric definition stable enough to compare years.

4. Conflict register

ID	Conflict	Parties	Why it is a real conflict
C1	Purpose of the score: management lever vs. protection from dismissal	S1 vs S2	F2 wants ranking and consequences; U1 forbids consequences. Cannot both be fully met.
C2	Driver-identified evidence for insurer vs. privacy	S3 vs S2/S5	I2 pushes for named, long-lived records; minimisation and U2 push the other way.
C3	Continuous tracking vs. off-duty privacy	S1 vs S2	F3 vs U2 on the same location data.
C4	Stable metric for insurer vs. tuning for fairness	S3 vs S2	I3 wants frozen thresholds; U4 demands thresholds correct for terrain and be fixable.
C5	Ranking / visibility of individual scores	S1 vs S2	F2 vs U2 on who sees whose score.
C6	Deviation alerts as control tool vs. legitimate detours	S1 vs S4	Road closures, rest stops and breakdowns generate false alerts; drivers fear being blamed.

5. Prioritisation method and results

Step 1 - Gate on the Musts (MoSCoW). Anything that breaches a Must-have constraint (lawful processing, no arbitrary dismissal, minimisation) is eliminated before scoring, however well it scores otherwise. This stops a high operational score from buying back a violation.

Step 2 - Weighted scoring of surviving options. Each option is scored 1-5 against four criteria (weights in brackets, total 100). Weights favour legal/ethical compliance because a breach would halt the programme (the union can refuse adoption; Legal can block release) and favour insurer evidence next because it funds the benefit.

Conflict	Option	Gate: breaches a Must?	Legal / ethical compliance (30)	Safety and insurer evidence value (25)	Operational value to fleet (25)	Driver / union acceptance (20)	Weighted /5
C1	A Score drives ranking and discipline	FAIL - eliminated	1	3	5	1	2.50
C1	B Score never used for anything	Pass	5	1	1	5	3.00
C1	C Advisory coaching score; discipline only via human-reviewed incident evidence	Pass	5	4	4	4	4.30
C2	A Named driver-level records to insurer	FAIL - eliminated	1	5	2	1	2.25
C2	B No data to insurer	Pass	5	1	1	4	2.80
C2	C Pseudonymised events + aggregates, independent audit right	Pass	4	4	3	4	3.75
C3	A Always-on location	FAIL - eliminated	1	3	5	1	2.50
C3	B Location only on request	Pass	4	1	1	5	2.70
C3	C Ignition-on only; off-duty privacy mode; 90-day raw purge	Pass	5	4	4	4	4.30
C4	A Thresholds fixed forever	Pass	3	5	3	1	3.10
C4	B Thresholds editable any time	Pass	3	1	3	5	2.90
C4	C Versioned metric, 12-month freeze, board-approved change with restated baseline	Pass	5	4	4	4	4.30
C5	A Public league table	FAIL - eliminated	2	2	5	1	2.55
C5	B Managers see no individual score	Pass	4	2	1	5	2.95

Conflict	Option	Gate: breaches a Must?	Legal / ethical compliance (30)	Safety and insurer evidence value (25)	Operational value to fleet (25)	Driver / union acceptance (20)	Weighted /5
C5	C Driver sees own; manager sees team distribution + individual scores only inside logged coaching access; no tables	Pass	5	4	4	4	4.30

Scores are the project team's judgement agreed in a prioritisation workshop; the value of the method is that the reasoning is visible and can be challenged. The winning option in every conflict is the "C" option, which is not a midpoint: it keeps the part of each side's need that is safe and drops the part that is not.

6. Resolution decisions - who gained, who gave up

ID	Decision	Fleet Manager	Union	Insurer
D1 (C1)	Score is an advisory coaching indicator . It can never be the sole or primary basis for dismissal, demotion, pay or discipline. Serious misconduct is judged on verified incident evidence by a human manager.	Gets coaching workflow and fleet KPIs. Gives up score-based discipline and rankings.	Gets the core assurance in writing (Data Use Policy, Section 6-7).	Neutral.
D2 (C2)	Insurer receives depot / terrain-class aggregates and pseudonymised event records; an independent auditor can verify raw samples under NDA. No driver names.	Neutral.	Identity protected from third parties.	Gets verifiable evidence. Gives up driver-level identity.
D3 (C3)	Telemetry only while ignition is on; off-duty privacy mode; raw 30-second data purged after 90 days.	Gives up always-on tracking; keeps in-shift tracking.	No off-duty tracking.	Neutral (events retained longer).
D4 (C4)	Insurer-facing KPI is the terrain-normalised harsh-braking rate per 100 km , v1.0 frozen for 12 months. Change only through the Governance Board, with baseline restated under the new rule.	Neutral.	Can still trigger a fix: terrain bug = emergency change with restatement.	Gets comparable metric. Gives up assumption of fixed thresholds if a fairness defect is found.
D5 (C5)	No league tables. Driver sees own score and breakdown; managers see team distribution, and individual scores only in logged coaching access.	Gives up public ranking.	Gets no shaming.	Neutral.
D6 (C6)	Deviation alerts are operational notices to the dispatcher with reason-code dismissal; they never feed the safety score or discipline. Known causes are suppressed.	Keeps timely alerts.	Drivers not blamed for detours.	Neutral.

7. Measurable terrain-normalisation requirements (the hill-route fix)

The earlier vendor judged braking with one flat threshold, so unavoidable braking on descents counted as bad driving. Fairness is therefore specified as **testable numbers**:

ID	Requirement	Measure / acceptance criterion
TN-1	Every GPS point is assigned a road gradient from an elevation model.	Gradient error ≤ 1.0 percentage point RMSE against surveyed check segments; missing gradient on $> 5\%$ of a driver's distance withholds the score instead of guessing.
TN-2	Harsh-braking threshold depends on downhill gradient band.	Net deceleration thresholds: 0 to under $2\% = 3.0 \text{ m/s}^2$; 2 to under $5\% = 3.3$; 5 to under $8\% = 3.6$; 8% and above $= 4.0$. Uphill and flat use 3.0 . Initial values, confirmed or re-tuned in the shadow-mode pilot, then versioned.
TN-3	Score is independent of route hilliness.	Spearman correlation between route hilliness (mean $ \text{gradient} $) and score $ \rho \leq 0.10$ on the validation cohort.
TN-4	Same driver, same behaviour, different terrain gives the same score.	Paired difference between flat-route and hill-route scores ≤ 2.0 points (cohort of ≥ 30 drivers who drive both).

ID	Requirement	Measure / acceptance criterion
TN-5	Genuinely harsh braking on hills is still penalised.	A 4.5 m/s ² event on a >=8% descent is counted in every run (no blanket leniency).
TN-6	Route reassignment must not move a score on its own.	Score change attributable to reassignment <= 2.0 points; no score used for action in the first 28 days on a new route.
TN-7	Release gate.	Release is blocked unless TN-3 to TN-5 pass and the hill-route regression cases in Deliverable 04 pass.

8. Residual disagreement and sign-off

- The Fleet Manager still prefers some visible ranking; this stays open and is revisited after the pilot via the Governance Board, not by quiet configuration change.
- The union asked for a veto on metric changes; the decision is a seat on the Governance Board plus the right to trigger an independent review (a veto was not accepted by the Fleet Manager).
- Sign-off required from: Fleet Manager, Union representative, Insurer liaison, Legal / Data Protection, Project Manager.